

Database Technologies – Assignment 8

- Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
Windows PowerShell
+-----+
| Tables_in_sales |
+-----+
| customers       |
| orders          |
| salespeople     |
+-----+
3 rows in set (0.013 sec)

KD4_98828_Gayatri@localhost>select * from orders
-> where cnum=(select cnum for customer where cname=cisneros)
-> ^C
KD4_98828_Gayatri@localhost>select * from orders
-> WHERE cnum=(SELECT cnum from customers WHERE cname='cisneros');
+-----+
| onum | amt   | odate   | cnum | snum |
+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+
2 rows in set (0.013 sec)

KD4_98828_Gayatri@localhost>select * from customers;
+-----+
| cnum | cname   | city    | rating | snum |
+-----+
| 2001 | Hoffman | London  | 100    | 1001 |
| 2002 | Giovanni | Rome    | 200    | 1003 |
| 2003 | Liu     | San Jose | 200    | 1002 |
| 2004 | Grass   | Berlin  | 300    | 1002 |
| 2006 | Clemens | London  | 100    | 1001 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
| 2007 | Pereira | Rome    | 100    | 1004 |
+-----+
7 rows in set (0.007 sec)

KD4_98828_Gayatri@localhost>select * from orders;
```

- Extract all the orders of Motika.
- Find all the orders of salespeople servicing customers in London.

```
Windows PowerShell
+-----+
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+
10 rows in set (0.008 sec)

KD4_98828_Gayatri@localhost>select * from salespeople;
+-----+
| snum | sname   | city    | comm |
+-----+
| 1001 | Peel    | London  | 0.12 |
| 1002 | Serres  | San Jose | 0.13 |
| 1004 | Motika  | London  | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York | 0.10 |
+-----+
5 rows in set (0.006 sec)

KD4_98828_Gayatri@localhost>select sname from salespeople where sname='motika';
+-----+
| sname |
+-----+
| Motika |
+-----+
1 row in set (0.007 sec)

KD4_98828_Gayatri@localhost>select * from orders
-> where snum=(select snum from salespeople where sname='motika');
+-----+
| onum | amt   | odate   | cnum | snum |
+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+
1 row in set (0.011 sec)

KD4_98828_Gayatri@localhost>select * from cu
```

- Write a query that selects all orders for amounts greater than any for the customers in London.

```

Windows PowerShell
+-----+
| sname |
+-----+
| Motika |
+-----+
1 row in set (0.007 sec)

KD4_98828_Gayatri@localhost>select * from orders
-> where snum=(select snum from salespeople where sname='motika');
+-----+
| onum | amt | odate | cnum | snum |
+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+
1 row in set (0.011 sec)

KD4_98828_Gayatri@localhost>select * from customers
-> where city='london';
+-----+
| cnum | cname | city | rating | snum |
+-----+
| 2001 | Hoffman | London | 100 | 1001 |
| 2006 | Clemens | London | 100 | 1001 |
+-----+
2 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM ORDERS
-> WHERE snum IN(SELECT SNUM FROM CUSTOMERS WHERE CITY='LONDON');
+-----+
| onum | amt | odate | cnum | snum |
+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+
3 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>

```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```

Windows PowerShell

KD4_98828_Gayatri@localhost>SELECT * FROM ORDERS
-> WHERE snum IN(SELECT SNUM FROM CUSTOMERS WHERE CITY='LONDON');
+-----+
| onum | amt | odate | cnum | snum |
+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+
3 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>SELECT snum FROM salespeople WHERE sname='serres';
+-----+
| snum |
+-----+
| 1002 |
+-----+
1 row in set (0.010 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM CUSTOMERS
-> WHERE RATING>=(SELECT snum FROM salespeople WHERE sname='serres');
Empty set (0.007 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM customers where rating >=(select rating from customers where snum IN(SELECT snum from salespeople WHERE sname='serres'));
ERROR 1242 (21000): Subquery returns more than 1 row
KD4_98828_Gayatri@localhost>SELECT * FROM customers
-> WHERE rating>ANY(SELECT RATING FROM customers WHERE snum IN (SELECT snum FROM salespeople WHERE sname='serres'));
+-----+
| cnum | cname | city | rating | snum |
+-----+
| 2002 | Giovanni | Rome | 200 | 1003 |
| 2003 | Liu | San Jose | 200 | 1002 |
| 2004 | Grass | Berlin | 300 | 1002 |
| 2008 | Cisneros | San Jose | 300 | 1007 |
+-----+
4 rows in set (0.011 sec)

KD4_98828_Gayatri@localhost>

```

11. Select customers who have a greater rating than any other customer in Rome.

```

Windows PowerShell
1 row in set (0.010 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM CUSTOMERS
-> WHERE RATING>=(SELECT snum FROM salespeople WHERE sname='serres');
Empty set (0.007 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM customers where rating >=(select rating from customers where snum IN(SELECT snum from sale
speople WHERE sname='serres'));
ERROR 1242 (21000): Subquery returns more than 1 row
KD4_98828_Gayatri@localhost>SELECT * FROM customers
-> WHERE rating>=ANY(SELECT RATING FROM customers WHERE snum IN (SELECT snum FROM salespeople WHERE sname='serres'));
+-----+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | Giovanni | Rome | 200 | 1003 |
| 2003 | Liu | San Jose | 200 | 1002 |
| 2004 | Grass | Berlin | 300 | 1002 |
| 2008 | Cisneros | San Jose | 300 | 1007 |
+-----+-----+-----+-----+-----+
4 rows in set (0.011 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM customers
-> WHERE rating > ANY( SELECT * FROM CUSTOMERS WHERE CITY='ROME');
ERROR 1241 (21000): Operand should contain 1 column(s)
KD4_98828_Gayatri@localhost>SELECT * FROM customers
-> WHERE rating>ANY(SELECT rating FROM CUSTOMERS WHERE CITY='ROME');
+-----+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | Giovanni | Rome | 200 | 1003 |
| 2003 | Liu | San Jose | 200 | 1002 |
| 2004 | Grass | Berlin | 300 | 1002 |
| 2008 | Cisneros | San Jose | 300 | 1007 |
+-----+-----+-----+-----+-----+
4 rows in set (0.008 sec)

KD4_98828_Gayatri@localhost>

```

12. Select all orders that had amounts that were greater that at least one of the orders from '1990-10-04'.

```

Windows PowerShell
ERROR 1241 (21000): Operand should contain 1 column(s)
KD4_98828_Gayatri@localhost>SELECT * FROM customers
-> WHERE rating>ANY(SELECT rating FROM CUSTOMERS WHERE CITY='ROME');
+-----+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | Giovanni | Rome | 200 | 1003 |
| 2003 | Liu | San Jose | 200 | 1002 |
| 2004 | Grass | Berlin | 300 | 1002 |
| 2008 | Cisneros | San Jose | 300 | 1007 |
+-----+-----+-----+-----+-----+
4 rows in set (0.008 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM ORDERS
-> WHERE amt>ANY(SELECT amt FROM ORDERS WHERE odate=1990-10-04);
Empty set, 1 warning (0.061 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM ORDERS
-> WHERE amt>ANY(SELECT amt FROM ORDERS WHERE odate='1990-10-04');
+-----+-----+-----+-----+-----+
| onum | amt | odate | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
8 rows in set (0.017 sec)

KD4_98828_Gayatri@localhost>

```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
Windows PowerShell
+-----+-----+-----+-----+
| 1001 | Peel | London | 0.12 |
| 1002 | Serres | San Jose | 0.13 |
| 1004 | Motika | London | 0.11 |
| 1007 | Rifkin | Barcelona | 0.15 |
| 1003 | Axelrod | New York | 0.10 |
+-----+-----+-----+-----+
5 rows in set (0.191 sec)

KD4_98828_Gayatri@localhost>select * from orders
-> where amt<(select city from salespeople where city='san jose');
Empty set (0.080 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM ORDERS
-> WHERE amt < ANY (SELECT city FROM customers where city='san jose');
Empty set (0.107 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM ORDERS
-> WHERE amt <ANY(select amt FROM orders WHERE cnum IN(SELECT cnum from customers where city='san jose'));
+-----+-----+-----+-----+
| onum | amt | odate | cnum | snum |
+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
+-----+-----+-----+-----+
8 rows in set (0.095 sec)

KD4_98828_Gayatri@localhost>
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
Windows PowerShell
+-----+-----+-----+-----+
2 rows in set (0.024 sec)

KD4_98828_Gayatri@localhost>SELECT o.amt, c.cname, c.rating
-> FROM orders o
-> JOIN customers c
-> ON o.cnum = c.cnum
-> WHERE o.amt > (
->     SELECT AVG(amt)
->     FROM orders
-> );
+-----+-----+-----+
| amt | cname | rating |
+-----+-----+-----+
| 5160.45 | Liu | 200 |
| 9891.88 | Clemens | 100 |
| 4723.00 | Clemens | 100 |
+-----+-----+-----+
3 rows in set (0.041 sec)

KD4_98828_Gayatri@localhost>
```

14. Select those customers whose rating are higher than every customer in London.

```
Windows PowerShell
| 3007 | 75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
+-----+-----+-----+-----+
8 rows in set (0.095 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM customer c
-> WHERE C.rating> IN(select cname from customers where c.city='london');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'IN(select cname from customers where c.city='london')' at line 2
KD4_98828_Gayatri@localhost>SELECT * FROM customer c
-> WHERE C.rating> (select cname from customers where c.city='london');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '-> WHERE C.rating> (select cname from customers where c.city='london')' at line 2
KD4_98828_Gayatri@localhost>^C
KD4_98828_Gayatri@localhost>^C
KD4_98828_Gayatri@localhost>SELECT * FROM CUSTOMERS
-> WHERE rating> ALL(select rating FROM customers where CITY ='LONDON');
+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+
| 2002 | Giovanni | Rome | 200 | 1003 |
| 2003 | Liu | San Jose | 200 | 1002 |
| 2004 | Grass | Berlin | 300 | 1002 |
| 2008 | Cisneros | San Jose | 300 | 1007 |
+-----+-----+-----+-----+
4 rows in set (0.063 sec)

KD4_98828_Gayatri@localhost>
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
Windows PowerShell
-> -> WHERE C.rating> (select cname from customers where c.city='london');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '-> WHERE C.rating> (select cname from customers where c.city='london')' at line 2
KD4_98828_Gayatri@localhost>^C
KD4_98828_Gayatri@localhost>^C
KD4_98828_Gayatri@localhost>SELECT * FROM CUSTOMERS
-> WHERE rating> ALL(select rating FROM customers where CITY ='LONDON');
+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+
| 2002 | Giovanni | Rome | 200 | 1003 |
| 2003 | Liu | San Jose | 200 | 1002 |
| 2004 | Grass | Berlin | 300 | 1002 |
| 2008 | Cisneros | San Jose | 300 | 1007 |
+-----+-----+-----+-----+
4 rows in set (0.063 sec)

KD4_98828_Gayatri@localhost>SELECT snum FROM salespeople
-> ^CERE city= ANY (SELECT CNUM FROM CUSTOMERS WHERE CITY=
KD4_98828_Gayatri@localhost>SELECT * FROM salespeople s
-> WHERE s.city <> ALL (SELECT c.city FROM CUSTOMERS c WHERE c.snum=s.snum);
+-----+-----+-----+-----+
| snum | sname | city | comm |
+-----+-----+-----+-----+
| 1004 | Motika | London | 0.11 |
| 1007 | Rifkin | Barcelona | 0.15 |
| 1003 | Axelrod | New York | 0.10 |
+-----+-----+-----+-----+
3 rows in set (0.074 sec)

KD4_98828_Gayatri@localhost>
```

9. Find names and numbers of all salesperson who have more than one customer.

```

Windows PowerShell
--> WHERE s.city <> ALL (SELECT c.city FROM CUSTOMERS c WHERE c.snum=s.snum);
+-----+-----+
| snum | sname | city | comm |
+-----+-----+
| 1004 | Motika | London | 0.11 |
| 1007 | Rifkin | Barcelona | 0.15 |
| 1003 | Axelrod | New York | 0.10 |
+-----+-----+
3 rows in set (0.074 sec)

KD4_98828_Gayatri@localhost>SELECT c.city FROM CUSTOMERS c WHERE c.snum=s.snum;
ERROR 1054 (42S22): Unknown column 's.snum' in 'where clause'
KD4_98828_Gayatri@localhost>SELECT c.city FROM CUSTOMERS C WHERE c.snum=s.snum;
ERROR 1054 (42S22): Unknown column 's.snum' in 'where clause'
KD4_98828_Gayatri@localhost>select sname,snum from salespeople
--> select sname,snum from salespeople
--> where snum IN(SELECT SNUM FROM CUSTOMERS GROUP BY SNUM HAVING COUNT(SNUM)>1);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to
use near 'select sname,snum from salespeople
where snum IN(SELECT SNUM FROM CUSTOMERS GRO' at line 2
KD4_98828_Gayatri@localhost>SELECT snum,sname
--> FROM salespeople
--> WHERE snum IN
--> (SELECT snum FROM customers
--> GROUP BY snum
--> HAVING COUNT(*) >1);
+-----+-----+
| snum | sname |
+-----+-----+
| 1001 | Peel |
| 1002 | Serres |
+-----+-----+
2 rows in set (0.094 sec)

KD4_98828_Gayatri@localhost>

```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```

Windows PowerShell
+-----+-----+
| 1001 | 15382.07 |
+-----+-----+
1 row in set (0.015 sec)

KD4_98828_Gayatri@localhost>SELECT s.snum, s.sname, s.city, COUNT(c.cnum)
AS customer_count
--> FROM salespeople s
--> JOIN customers c
--> ON s.snum = c.snum
--> GROUP BY s.snum, s.sname, s.city
--> HAVING COUNT(c.cnum) > 1;
+-----+-----+-----+-----+
| snum | sname | city | customer_count |
+-----+-----+-----+-----+
| 1001 | Peel | London | 2 |
| 1002 | Serres | San Jose | 2 |
+-----+-----+-----+-----+
2 rows in set (0.024 sec)

KD4_98828_Gayatri@localhost>

```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
Windows PowerShell
KD4_98828_Gayatri@localhost>SELECT c.city FROM CUSTOMERS C WHERE c.snum=s.snum;
ERROR 1054 (42S22): Unknown column 's.snum' in 'where clause'
KD4_98828_Gayatri@localhost>select sname,snum from salespeople
-> select sname,snum from salespeople
-> where snum IN(SELECT SNUM FROM CUSTOMERS GROUP BY SNUM HAVING COUNT(SNUM)>1);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to
use near 'select sname,snum from salespeople
where snum IN(SELECT SNUM FROM CUSTOMERS GRO' at line 2
KD4_98828_Gayatri@localhost>SELECT snum,sname
-> FROM salespeople
-> WHERE snum IN
-> (SELECT snum FROM customers
-> GROUP BY snum
-> HAVING COUNT(*) >1);
+-----+-----+
| snum | sname |
+-----+-----+
| 1001 | Peel  |
| 1002 | Serres |
+-----+-----+
2 rows in set (0.094 sec)

KD4_98828_Gayatri@localhost>SELECT snum, SUM(amt) AS total_amount
-> FROM orders
-> GROUP BY snum
-> HAVING SUM(amt) > (
-> SELECT MAX(amt)
-> FROM orders
-> );
+-----+-----+
| snum | total_amount |
+-----+-----+
| 1001 | 15382.07 |
+-----+-----+
1 row in set (0.015 sec)

KD4_98828_Gayatri@localhost>
```